

[2-16:] The sum of all minterms of a Boolean function of n variables is 1.

(a) Prove the above statement for $n=3$.

Solution: Let the Boolean function be $F(A, B, C)$.

The minterms for $F(A, B, C)$ are: $A'B'C'$, $A'B'C$, $A'BC'$, $A'BC$, $AB'C'$, $AB'C$, ABC' , ABC .

ABC

$$A'B'C' + A'B'C = A'B'(C' + C) \text{ (Postulate 4(a))}$$

$$= A'B'(C + C') \text{ (Postulate 3(a))}$$

$$= A'B' \cdot 1 \text{ (Postulate 5(a))}$$

$$= A'B' \text{ (Postulate 2(b))}$$

$$A'BC' + A'BC = A'B(C' + C) \text{ (Postulate 4(a))}$$

$$= A'B(C + C') \text{ (Postulate 3(a))}$$

$$= A'B \cdot 1 \text{ (Postulate 5(a))}$$

$$= A'B \text{ (Postulate 2(b))}$$

$$A'BC' + ABC' = AB'(C' + C) \text{ (Postulate 4(a))}$$

$$= AB'(C + C') \text{ (Postulate 3(a))}$$

$$= AB' \cdot 1 \text{ (Postulate 5(a))}$$

$$= AB' \text{ (Postulate 2(b))}$$

$$ABC' + ABC = AB(C' + C) \text{ (Postulate 4(a))}$$

$$= AB(C + C') \text{ (Postulate 3(a))}$$

$$= AB \cdot 1 \text{ (Postulate 5(a))}$$

$$= AB \text{ (Postulate 2(b))}$$

$$A'B' + A'B = A'(B' + B) \text{ (Postulate 4(a))}$$

$$= A'(B + B') \text{ (Postulate 3(a))}$$

$$= A' \cdot 1 \text{ (Postulate 5(a))}$$

$$= A' \text{ (Postulate 2(b))}$$

$$AB' + AB = A(B' + B) \text{ (Postulate 4(a))}$$

$$= A(B + B') \text{ (Postulate 3(a))}$$

$$= A \cdot 1 \text{ (Postulate 5(a))}$$

$$= A \text{ (Postulate 2(b))}$$

$$A' + A = A + A' \text{ (Postulate 3(a))} = 1 \text{ (Postulate 5(a))}$$